

feasibility

Aurora-OS / Harmony-OS client recording — autonomous-path feasibility (HXC-108 residual)

Date: 2026-06-23 **Author:** research subagent (read-only + web research) **Mandate:** §11.4.150 multi-angle deep research BEFORE classifying the Aurora/Harmony recording gaps. Determine whether ANY autonomous path exists to record the HelixCode Aurora-OS and Harmony-OS Fyne clients genuinely running, or whether they are genuine §11.4.98(B) operator-bootstrap blockers (and not §11.4.112 structurally-impossible).

Confirmed host / repo facts (FACT — captured this session)

- **macOS host:** arm64, **Apple M3 Pro**, Apple Silicon (hw.optional.arm64=1). Darwin 24.5.
 - **macOS toolchains ABSENT:** sfdk mb2 sb2 rpmbuild (Aurora) and hdc ohpm devecos hvigow (Harmony) all NOT FOUND on this host. (Mission-confirmed; re-verified.)
 - **thinker.local (FACT, probed read-only this session):** reachable, Linux thinker 6.17.0-23-generic ... Ubuntu 24.04 ... x86_64 (KVM host). **NONE** of sfdk mb2 sb2 rpmbuild hdc ohpm devecos hvigow are installed there either (each NOTFOUND). → Neither SDK happens to be pre-installed on the Linux box. (Another agent owns thinker for Android; only a single read-only command -v probe was run.)
 - **The apps are Fyne Go clients (FACT):** applications/aurora_os and applications/harmony_os both import fyne.io (in main.go + theme.go). Build targets exist: make aurora-os → bin/aurora-os, make harmony-os → bin/harmony-os (Makefile L599–609). They cross-compile but need the vendor SDK + a target device/emulator to RUN.
-

Angle 1 — Aurora OS SDK on Linux + emulator headless feasibility

FACT. Aurora OS (OC Аврора, Open Mobile Platform / OMP — a Sailfish-derived Russian mobile OS) ships an officially **Linux-installable SDK** (“Аврора SDK”, incl. the MB2 build engine). Evidence: - Linux download page exists: `developer.auroraos.ru/downloads/sdk_mb2/5.1.3.85/linux` (also Windows + macOS variants). - The SDK installs **two VirtualBox VMs** — a **Build Engine** VM (SSH-driven, host-OS- independent builds via `mb2 -t AuroraOS-armv7hl build / AuroraOS-i486`) and an **Emulator** VM (runs the app “similar to a real device”). - Host requirements: Linux/Win/macOS, ~5 GB disk, 4 GB RAM, hardware virtualization recommended, **Oracle VM VirtualBox required and pre-installed**. - There is an emulator-management doc (“Управление эмуляторами”) and an `sfdk-class` CLI; a community `aurora-cli` (keygenqt) wraps SDK/emulator ops, and `keygenqt/aurora-m1` documents running the Aurora emulator on Apple Silicon via **VirtualBox + VNC**.

Headless / autonomy assessment (FACT + UNCONFIRMED): - The emulator is a **GUI VirtualBox VM**, not a headless CLI surface. Capturing the client running means screen-capturing the emulator window (a \$11.4.117 pixel/OCR oracle), not reading a terminal. (FACT: it is a VM with a graphical framebuffer.) - Running it requires **VirtualBox on the host**. On the **Apple-Silicon macOS host this is a hard blocker** — VirtualBox on Apple Silicon is experimental/unreliable; the community path (`aurora-m1`) is non-trivial bootstrap. (FACT: VirtualBox is x86-oriented; UNCONFIRMED whether a working Apple-Silicon VirtualBox + Aurora emulator combo is reproducible autonomously.) - On **thinker (Linux x86_64)** VirtualBox could host the VMs, BUT it is a **KVM host** — running VirtualBox VMs alongside/under KVM hits nested-virtualization contention, and the box is **owned by the Android agent** (§11.4.119 single-resource-owner). (UNCONFIRMED that VirtualBox + Aurora emulator runs cleanly there without disturbing Android work.) - **Access gate:** the SDK lives behind the Russian developer portal `developer.auroraos.ru`; download/registration may be **geo/account-gated** (§11.4.3 `geo_restricted` risk — UNCONFIRMED whether anonymous download works from this locale).

Sources: `developer.auroraos.ru` SDK MB2 Linux download + `setup_linux` + `manage_emulators` docs; `omp.ru/sdk`; `keygenqt aurora-cli` + `aurora-m1` (URLs in footer).

Angle 2 — Harmony OS / DevEco on Linux + emulator path

FACT. DevEco Studio (the IDE) is Windows + macOS ONLY — no Linux build. Linux gets only **command-line tools** (hvmigor, hdc) for build/sign/deploy — not an emulator. - **hdc** (HarmonyOS Device Connector) runs on Win/Linux/macOS — but it only *connects to* a device/emulator; it does not *provide* one. - **The HarmonyOS NEXT emulator does NOT run on Linux**, and one authoritative-community source states it “only runs on Windows and won’t work through ANY VM” (needs Hyper-V + 12 GB RAM). (FACT for Windows; see contradiction below for macOS.) - **Conflicting-but-favourable data (FACT, latest source):** DevEco Studio 5.0.3 Beta2 (API 15) for macOS ships a “100% functional” emulator with **Apple-Silicon support** (Rosetta-assisted). The official site historically gave **no emulator for Intel Macs** — so the macOS emulator path is **Apple-Silicon-specific**. *This host is Apple M3 Pro*, so the macOS-Apple-Silicon emulator is technically applicable here. - **Hard credential gate:** downloading the Command-Line Tools AND the SDK **requires a registered + verified Huawei Developer account login**; signing-key generation also requires that account. (FACT — credential bootstrap, §11.4.10 / §11.4.98(B) exception. Account verification is frequently China-region-gated — §11.4.3 geo_restricted risk, UNCONFIRMED for this locale.)

Autonomy assessment: even on the favourable macOS-Apple-Silicon path, RUN requires: (1) installing the **GUI DevEco Studio** + SDK (absent here), (2) a **verified Huawei account login** (operator credential, possibly geo-gated), (3) launching the emulator **through the GUI IDE** (not a headless CLI), then (4) a §11.4.117 pixel/OCR window capture of the Fyne client running on it. None of these are blocked by platform *design* — they are toolchain-install + vendor-account + GUI-driving gates.

Sources: developer.huawei.com DevEco docs (ide-tools-overview, ide-software-install); harmony-developers.com (5.0.3 Beta2 macOS emulator); device.harmonyos.com local-emulator FAQ; Servo Book OpenHarmony page; XDA NEXT-emulator thread; Wikipedia DevEco Studio (URLs in footer).

Angle 3 — emulator-in-container precedents

FACT/UNCONFIRMED. Android has the well-known android-emulator-container-scripts (headless emulator in a container, ADB/gRPC + VNC/WebRTC streaming) — the clean precedent the Android agent leverages. **No equivalent first-class container-emulator exists for either Aurora or Harmony:** - Aurora’s emulator is a **VirtualBox VM** — containerising it means nested virt (heavy, fragile); no published turnkey “Aurora emulator in a container” project found (the closest is the VirtualBox+VNC manual path). - Harmony’s NEXT emulator is reported to **refuse ANY VM** (Windows/Hyper-V case) and is otherwise the macOS-

Apple-Silicon native emulator inside the GUI IDE — **no container path found**.
(FACT: no precedent surfaced in search; UNCONFIRMED that none exists.)

Angle 4 — Is the Fyne Go client on a generic desktop a legitimate §11.4.158 proxy?

Honest boundary (FACT — NO). The HelixCode Aurora/Harmony clients are Fyne apps; make desktop builds the *same shared Fyne UI* for Linux/macOS. Recording that desktop build would prove the **shared Fyne UI logic renders + the LLM-core wiring works**, but it is a **different artifact running on a different OS** — it does **NOT** exercise the Aurora or Harmony **native target** (different cross-compile target, vendor windowing/IME, platform integration). Per §11.4.143 (real user journey on the real target) and §11.4.158 (every feature recorded genuinely working on its platform), a Fyne-on-Linux/macOS recording is **at most a supplementary UI smoke test, NOT a substitute** for “the Aurora/Harmony client genuinely running.” Claiming it as the platform recording would be a §11.4 PASS-bluff at the platform-target layer. (This is the same honest boundary the mission flags.)

Angle 5 — thinker.local SDK read-probe

FACT (this session): `ssh thinker.local 'command -v sfdk mb2 sb2 rpmbuild hdc ohpm deveco hvigorw'` → every tool NOTFOUND. thinker is a clean Ubuntu 24.04 x86_64 KVM host with neither SDK installed. No autonomous shortcut via a pre-provisioned Linux box.

Per-platform VERDICT

Aurora OS → operator-bootstrap-required (§11.4.98(B)) — NOT structurally-impossible

A real autonomous path is *technically* describable (Linux x86_64 + VirtualBox + Aurora SDK → emulator VM → §11.4.117 pixel/OCR capture), so §11.4.112 **does NOT apply**. But every prerequisite is an operator-bootstrap / credential / resource gate, none reachable autonomously from the current state: 1. Install Aurora SDK (absent on both macOS host and thinker) + **VirtualBox**. 2. SDK is behind the OMP/developer.auroraos.ru portal — likely **account/geo-gated** (§11.4.3 risk, UNCONFIRMED). 3. VirtualBox on **Apple Silicon** = unreliable; on **thinker** = nested-virt under KVM + the box is **owned by the Android agent** (§11.4.119) → not an available resource now.

Recommended classification: Operator-blocked (§11.4.21) with enumerated unblock choices (§11.4.148-D3): **[A]** operator provisions a dedicated **x86_64 Linux box with VirtualBox + Aurora SDK** (account supplied) → agent then drives emulator + §11.4.117 window-scoped recording; **[B]** operator provides Aurora SDK creds + authorises shared use of an x86_64 host → same; **[C]** defer as documented §11.4.98(B) operator-bootstrap gap with this research as the cited evidence the path was exhausted.

Harmony OS → operator-bootstrap-required (§11.4.98(B)) — NOT structurally-impossible

The macOS-Apple-Silicon DevEco emulator (5.0.3 Beta2+) means a RUN path **does exist on a host of this exact class** (Apple M3 Pro), so **§11.4.112 does NOT apply**. Blockers are bootstrap/credential/GUI gates: 1. Install **GUI DevEco Studio + SDK** (absent here; no Linux IDE exists). 2. **Verified Huawei Developer account login** required for SDK + signing + emulator (operator credential, §11.4.10; verification often China-region-gated — §11.4.3 risk, UNCONFIRMED for this locale). 3. Emulator launches **through the GUI IDE** → drive + §11.4.117 pixel/OCR window capture.

Recommended classification: Operator-blocked (§11.4.21) with unblock choices: **[A]** operator installs DevEco Studio on this Apple-Silicon host + completes Huawei verified-account login → agent drives the IDE emulator + window-scoped recording (§11.4.159); **[B]** operator supplies a physical HarmonyOS device + Huawei account → hdc-driven on real hardware; **[C]** defer as documented §11.4.98(B) gap with this research cited.

Honest recommendation

Neither platform is **structurally-impossible** (§11.4.112 is **NOT** warranted — a credible autonomous run path exists for each once the toolchain + vendor account are present), and neither has an autonomous path **available from the current host state** (SDKs absent everywhere probed; vendor-account + GUI-IDE/VirtualBox + possible geo gates). They are genuine **§11.4.98(B) operator-bootstrap blockers**. The §11.4.150/§11.4.8 deep-research path is hereby **exhausted and documented**; the correct closure is **Operator-blocked (§11.4.21)** per platform with the enumerated unblock choices above — **not** a faked recording, **not** a Fyne-on-desktop proxy passed off as the native target (§11.4.143/ §11.4.158 forbid it), and **not** a structurally-impossible won't-fix. Residual UNCONFIRMED items an operator can resolve cheaply: (i) anonymous vs account/geo-gated Aurora SDK download from this locale; (ii) Huawei verified-account availability/region from this locale; (iii) reproducibility of VirtualBox+Aurora-emulator on Apple Silicon vs an x86_64 Linux box.

Sources verified 2026-06-23

- Aurora SDK MB2 Linux download — https://developer.auroraos.ru/downloads/sdk_mb2/5.1.3.85/linux
- Aurora SDK Linux setup — https://developer.auroraos.ru/doc/5.2.0/sdk/app_development/setup/setup_linux
- Aurora emulator management — https://developer.auroraos.ru/doc/sdk/app_development/setup/manage_emulators
- Aurora Scratchbox2 / mb2 — <https://developer.auroraos.ru/doc/sdk/tools/scratchbox2>
- OMP Aurora SDK — <https://www.omp.ru/sdk>
- Aurora CLI (keygenqt) — <https://keygenqt.github.io/aurora-cli/>
- Aurora on Apple Silicon (VirtualBox+VNC) — <https://github.com/keygenqt/aurora-m1>
- HUAWEI DevEco tools overview — <https://developer.huawei.com/consumer/en/doc/harmonyos-guides/ide-tools-overview>
- HUAWEI DevEco install — <https://developer.huawei.com/consumer/en/doc/harmonyos-guides/ide-software-install>
- DevEco macOS emulator (5.0.3 Beta2, Apple Silicon) — <https://www.harmony-developers.com/p/global-developers-can-now-download>
- HarmonyOS local-emulator FAQ — <https://device.harmonyos.com/en/docs/apiref/doc-guides/faq-local-emulator-0000001116085454>
- OpenHarmony build / CLI tools (Servo Book) — <https://book.servo.org/building/openharmony.html>
- DevEco Studio — https://en.wikipedia.org/wiki/DevEco_Studio
- Android emulator-container precedent — <https://github.com/google/android-emulator-container-scripts>